

[BUG] QAOA layer template does not work with TensorFlow

<https://github.com/PennyLaneAI/pennylane/issues/1611>

ROW 3

[BUG] QAOA layer template does not work with TensorFlow #1611

Closed

#1615



licedric opened on Aug 31, 2021

Expected behavior

PennyLane's QAOA layer templates should work with the TensorFlow interface.

Actual behavior

A `TypeError: object of type 'ResourceVariable' has no len()` is thrown.

Additional information

No response

Source code

```
import networkx as nx
import pennylane as qml
import numpy as np
import tensorflow as tf

num_nodes = 4
num_edges = 4
p = 2
seed = 1967

def qaoa_layer(gamma, alpha):
    qml.qaoa.cost_layer(gamma, cost_h)
    qml.qaoa.mixer_layer(alpha, mixer_h)

def circuit(gamma, alpha, **kwargs):
    for i in range(num_nodes):
        qml.Hadamard(wires=i)

    qml.layer(qaoa_layer, p, gamma, alpha)

g = nx.gnm_random_graph(num_nodes, num_edges, seed=seed)
cost_h, mixer_h = qml.qaoa.maxcut(g)

dev = qml.device('default.qubit', wires=4)

alpha = tf.Variable(np.random.rand(p), dtype=tf.float64)
gamma = tf.Variable(np.random.rand(p), dtype=tf.float64)
cost_function = qml.ExpvalCost(circuit, cost_h, dev, interface='tf', optimize=True)
cost_function(gamma, alpha)
```

Tracebacks

```
-----
TypeError                                Traceback (most recent call last)
/var/folders/v1/y849xjw53gl8_4b7cf9qr4pxc7s0j7/T/ipykernel_1781/1824392795.py in <module>
----> 1 cost_function(gamma, alpha)

~/pyenv/versions/3.9.6/envs/braket-dev-3.9.6/lib/python3.9/site-packages/pennylane/vqe/vqe.py in __call__(self, *args, **kwargs)
   585
   586     def __call__(self, *args, **kwargs):
--> 587         return self.cost_fn(*args, **kwargs)
   588
   589

~/pyenv/versions/3.9.6/envs/braket-dev-3.9.6/lib/python3.9/site-packages/pennylane/vqe/vqe.py in cost_fn(*args, **kwargs)
   575         total = 0
   576         for o, c in zip(obs_groupings, coeffs_groupings):
--> 577             res = circuit(*qnode_args, obs=o, **qnode_kwargs)
   578             total += sum([r * c_ for r, c_ in zip(res, c)])
   579         return total
```

Microsoft

```

~/pyenv/versions/3.9.6/envs/braket-dev-3.9.6/lib/python3.9/site-packages/pennylane/qnode.py in __call__(self,
593         if self.mutable or self.qtape is None:
594             # construct the tape
--> 595             self.construct(args, kwargs)
596
597             # execute the tape

~/pyenv/versions/3.9.6/envs/braket-dev-3.9.6/lib/python3.9/site-packages/pennylane/qnode.py in construct(self,
522
523         with self.qtape:
--> 524             self.qfunc_output = self.func(*args, **kwargs)
525
526             if not isinstance(self.qfunc_output, Sequence):

~/pyenv/versions/3.9.6/envs/braket-dev-3.9.6/lib/python3.9/site-packages/pennylane/vqe/vqe.py in circuit(obs,
568         def circuit(*qnode_args, obs, **qnode_kwargs):
569             """Converting ansatz into a full circuit including measurements"""
--> 570             ansatz(*qnode_args, wires=w, **qnode_kwargs)
571             return [qml.expval(o) for o in obs]
572

/var/folders/v1/y849xjw53gl8_4b7cf9qr4pxc7s0j7/T/ipykernel_1781/926742614.py in circuit(gamma, alpha, **kwargs)
3         qml.Hadamard(wires=i)
4
----> 5         qml.layer(qaoa_layer, p, gamma, alpha)

~/pyenv/versions/3.9.6/envs/braket-dev-3.9.6/lib/python3.9/site-packages/pennylane/templates/decorator.py in
59
60         with qml.tape.OperationRecorder() as rec:
--> 61             func(*args, **kwargs)
62
63             return rec.queue

~/pyenv/versions/3.9.6/envs/braket-dev-3.9.6/lib/python3.9/site-packages/pennylane/templates/layer.py in laye
211         """
212
--> 213         _preprocess(args, depth)
214
215         for i in range(0, int(depth)):

~/pyenv/versions/3.9.6/envs/braket-dev-3.9.6/lib/python3.9/site-packages/pennylane/templates/layer.py in _pre
31         for arg in args:
32
--> 33             if len(arg) != depth:
34                 raise ValueError(
35                     "Each positional argument must have length matching 'depth'; expected {} got {}".format

TypeError: object of type 'ResourceVariable' has no len()

```

System information

```

Name: PennyLane
Version: 0.16.0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author: None
Author-email: None
License: Apache License 2.0
Location: /Users/licedric/.pyenv/versions/3.9.6/envs/braket-dev-3.9.6/lib/python3.9/site-packages
Requires: scipy, autoray, autograd, appdirs, networkx, toml, semantic-version, numpy
Required-by: PennyLane-Qchem, amazon-braket-pennylane-plugin
Platform info:          macOS-10.15.7-x86_64-i386-64bit
Python version:         3.9.6
Numpy version:          1.19.5
Scipy version:           1.7.0
Installed devices:
- default.gaussian (PennyLane-0.16.0)
- default.mixed (PennyLane-0.16.0)
- default.qubit (PennyLane-0.16.0)

```



 **licedric** added **bug** on Aug 31, 2021

 **github-actions** assigned [antalszava](#) on Aug 31, 2021



josh146 on Aug 31, 2021

Member ...

Thanks [@licedric](#)! From looking into this further, it appears that the issue is that `qml.layer` uses `len(args)`, which will not work if any of the input arguments to be layered are `tf.Variable` objects which are not iterable.

A solution would be to either:

- Avoid using `len()`, by replacing it with `qml.math.shape(args)[0]`
- Convert `args` to a `tf.Tensor` (which *is* iterable) if it happens to be a `tf.Variable`.

The first option is preferable, since it is framework agnostic.

We should also add TF and Torch tests for `qml.layer`.



 **albi3ro** assigned [albi3ro](#) and unassigned [antalszava](#) on Aug 31, 2021

 **albi3ro** mentioned this on Aug 31, 2021

 [qml.layer support for tensorflow Variable #1615](#)

 **albi3ro** closed this as **completed** in [#1615](#) on Aug 31, 2021

<> Code ▾

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qml.layer support for tensorflow Variable #1615

 Merged albi3ro merged 7 commits into `master` from `bugfix-layer-tf` on Aug 31, 2021

Conversation 4

Commits 7

Checks 0

Files changed 3



albi3ro commented on Aug 31, 2021

Contributor



Fixes [#1611](#)

`qml.layer` used `len` to validate the length of the input argument, but tensorflow `tf.Variable` objects do not support calculation of `len`. Instead we use `qml.math.shape(arg)[0]` to allow the template to support all interfaces.



  fix layer with tensorflow bug 47a05e7

  albi3ro added the `bug` label on Aug 31, 2021

  Merge branch 'master' into bugfix-layer-tf 389e57e

 albi3ro commented on Aug 31, 2021

[View reviewed changes](#)

`.github/CHANGELOG.md`

 Show resolved

  Update `.github/CHANGELOG.md` 53db3cd